FairCOVENANT: Smart Trade Hubs to Accelerate Fair Global Trade

Building Trusted City-to-City Trade Corridors for the Digital Era

> "Micro, that is small and medium-sized enterprises (MSMEs) are, quite literally, the foundations of the global economy. They are engines of economic growth and employment, **accounting for 90% of businesses, up to 70% of all jobs and 50% of gross domestic product (GDP)** at the global level." [1]

Unleashing the growth potential of micro, small and medium enterprises (MSMEs) is the key to the next wave of global development. Enabling MSMEs to participate in global trade is a key part of this strategic opportunity. That means reducing barriers to MSME participation in global trade, streamlining cross-border transactions processes and costs, and mitigating geopolitical risks.

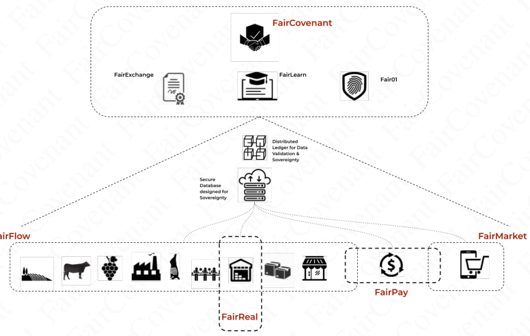
Buyers and **sellers** need to be able to access each other with minimal leakages of value via unnecessary intermediaries. Border regulators, such as customs, need low-cost capabilities to vet goods coming and going. Confidence in the integrity of the product manufacturers and service providers is vital - for all concerned. Having access to **credible payments, settlements, and insurance products** and services is fundamental to enabling participation.

MSMEs can also overcome the constraints of scale, through **virtual aggregation** just as buyers can access diverse supply options with reduced search, validation and transaction costs.

Underpinning all of this is information infrastructure that enables end-to-end supply chain orchestration to be mobilized to connect buyers and sellers and move goods through customs and ports. Digitization helps unlock this potential, but must be done in credible ways. Data integrity systems, embedded into the physical fabric of supply chains and key logistics nodes, enable accelerated trade growth. Sister city relations provide the overarching collaborative frame to activate at scale.

The FairCOVENANT **Smart Trade Hubs** initiative leverages existing technologies to drive a high value proposition that can scale to global application.

FairCOVENANT: SMART TRADE HUBS



COVENANT: A solemn binding agreement. This idea underpins our approach to ecosystem development through the integration of digital technologies with real-life infrastructure and institutions aimed at enabling functional, low-cost, high-value interactions.

FAIR: Impartial and just, without favoritism or discrimination; without cheating or trying to achieve unjust advantage. For trade to be sustainable, it must be fair for all involved.

Key Terminology

- Fair - this is the overarching name for the integrated project initiative.
- Fair01 - this refers to the AI enabled data oracle that powers real-time validation of data to reduce supply chain performance risks, and enables financial innovation such as payments, credit, and insurance, that streamline payments.
- FairLearn - this refers to our training and incubation initiative that can support the development of local talents who support applications and innovations.
- FairExchange - this refers to the foundational "rules" and "protocols" that define data validation for cross-

[1] United Nations (26 June 2023) Micro-, Small and Medium-sized Enterprises Are Key to an Inclusive and Sustainable Future. Accessed: `https://tinyurl.com/584dywjz`

border digitalization-enabled trade purposes. Non-technical institutions such as sister city agreements are critical "real world" anchors to be leveraged.

- FairFlow - this refers to our existing supply chain digitalization, orchestration and trade documentation application with cloning capabilities to support localized deployment and customization.
- FairMarket - this refers to the e-commerce environment that is seamlessly integrated with FairFlow.
- FairPay - this refers to the module of payments and financial solutions interfaces that can be expanded to support solutions innovation.
- FairReal - this refers to the "real economy" foundations of the ecosystem, and includes warehousing, logistics, cold storage, industrial zones, and trade zones etc.

1. Introduction

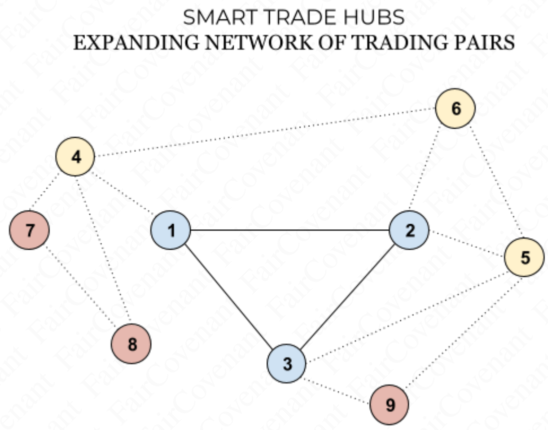
FairCOVENANT Smart Trade Hubs is a city-to-city trade accelerator, designed to create trusted trade corridors between sister cities, empowering micro, small, and medium enterprises (MSMEs) to access global markets.

Smart Trade Hubs can fan out as a network of integrated trading ecosystems, connecting businesses across the globe via credible platforms that are built on the foundations of data integrity in supply chains.

The initiative described in this paper leverages existing technologies, developed through years of university-based research and government support in Australia, Africa, and elsewhere. This means we are able to "hit the ground running" while also mapping out a development roadmap that enables scaling and leveraging the potential of new technologies such as Artificial Intelligence (AI) and machine learning. The Smart Trade Hubs initiative is a model that integrates:

- **Physical trade infrastructure**: wholesale display, warehousing and logistics.
- **Supply chain digitalization backbone**: enabling distributed orchestration, streamlined use by MSMEs, and virtual aggregation benefits for buyers and sellers.
- **Digital governance layers**: for data integrity and compliance validation.
- **AI-driven verification oracles**: to ensure real-time transaction trust.
- **Payments and insurance solutions**: to empower those previously unable to access these services; and
- **A FairLearn Center**: to develop local digital talent for long-term sustainability.

Our first demonstration case will leverage a proven trade corridor model - **anchored by sister city relations** - and modernize it with AI-enabled compliance systems, Halal certification assurance where applicable, [2] and digital finance capabilities. To do this, we will deploy and expand existing digital capabilities - our FairFlow and FairMarket Apps. We then extend out with additional hub-to-hub pairs, progressively building a network of interconnected Smart Trade Hubs.



This initiative will serve as a real-world living laboratory, producing a replicable blueprint for scaling across regions and institutionalizing global standards. It therefore creates extensive scaling opportunities for early participants.

Why now?

1. Global growth is tepid at best. New engines of growth need to be mobilized. MSMEs are locked out of global trade due to lack of counterparty trust, high compliance costs, high transaction costs (often associated with multilayered intermediary structures) and poor access to finance. [3] Yet, in most economies, MSMEs are the employment backbones whose full value potential remains untapped. Between 70-80% of employment in developing economies are provided by MSMEs, and about 60% in developed countries. [4] MSMEs contribute about 50% to global GDP.
2. Global buyers demand verified legitimacy and quality in complex supply chains, which is documentary or data-intensive and backed by robust financial support (credit and insurance). This requirement has historically locked out MSMEs, even if larger intermediaries ultimately rely on a network of smaller firms to deliver the final products or services.
3. Trade finance and insurance require real-time data and real-time data[5] integrity for risk reduction [6] Data integrity in a document-centric environment remains clouded in risks associated with identity and documen-

[2] The Halal Secure platform has already been developed for incorporation into this initiative. See: https://halalsecure.com/

[3] See World Economic Forum (May 2025) How small businesses can navigate global trade in an era of polycrisis. Accessed: https://tinyurl.com/bdusud3j

[4] McKinsey (2024) A microscope on small businesses. Accessed at https://tinyurl.com/4y6jst6f

[5] See Emmanuelle Ganne and Kathryn Lundquist (World Trade Organization) (2019) "The Digital Economy, GVCs and SMEs." Accessed here: https://tinyurl.com/ms9h8e3m

[6] See World Trade Organization (2016). Trade obstacles to SME participation in trade. Accessed at: https://tinyurl.com/57turmhz

tary fraud, not to mention problems associated with the economics of large and growing bodies of data.

4. Digitalization can be applied to both production efficiency and productivity growth as well as to process or transactions-related productivity. Our focus on transaction-process solutions taps into, and enables the release of value potential through reduced information costs, enhanced coordination, lower finance risks and costs and accelerated flow of goods and payments.

The Fair Smart Trade Hubs initiative addresses these gaps through a scaffolded bottom-up implementation, creating safe and fair trade channels that combine physical trade hubs, digital supply chain orchestration and documentation integrity systems, AI governance, and sister-city partnerships.

Call to Action: We seek municipal and/or regional government anchors, industrial and trade zone operators, warehousing partners, AI foundational model developers, and finance/insurance partners to co-create this pilot and capture first-mover advantages in shaping the next generation of global trade infrastructure.

2. The Problem

Despite technology advances, MSMEs remain disadvantaged in global trade:

1. **High Transaction Costs**: buyers and sellers experience high costs in finding suitable transaction partners. Fragmented supply chains drive enterprises towards intermediaries. Furthermore, fragmented supply chains limit MSMEs from access or developing value-adding capabilities to grow local industries.
2. **Trust Deficit**: Buyers face uncertainty about supplier legitimacy, product quality, and compliance with standards. Put simply, there is no visibility through the supply chain, and finding potential suppliers or credible buyers is time and resource intensive. Data deficits are central to the absence of trustworthy participation in supply chains by MSMEs.
3. **Fragmented Compliance**: Regulatory processes differ across jurisdictions, creating friction and cost. A clear example is in the area of Halal market access. Halal trade requires rigorous compliance, but current certification systems lack digital integrity and interoperability. The global Halal market was estimated at $2.71 trillion in 2024 and is projected to reach $5.91 trillion by 2033, growing at a compound annual growth rate (CAGR) of about 9.0%, and is a clear sector of opportunity and need.
4. **Finance Access Gap**: Traditional trade finance avoids MSMEs due to unverifiable trade data and perceived risk. Similarly, the inability to access cost-effective insurance support makes compliant supply all but impossible for many MSMEs. They, and their products are in effect, “the un-insured.”

At the same time global buyers seek secure supply chains in an era of risk and disruption while governments aim to modernize trade facilitation without losing sovereignty. This means confidence in the source of supplies and the ability to preserve data security and comply with data sovereignty requirements. These needs are increasingly significant as nations address concerns about national security dimensions associated with data storage and distribution.

Current models - platform-only solutions like Alibaba and eBay, with opaque supply chain networks - fail to address the trust and compliance challenge for regulated, high-value trade. The result is persistence of vendor impersonation, [7] product [8] and payment fraud (which further undermines system trust), [9] and uneconomic finance and insurance, which ultimately excludes MSMEs.

Furthermore, MSMEs struggle with trade-related documentation, which acts as a further barrier to connecting with global markets and the ability to distribute directly to customers around the world. Moves towards paperless trade continue to progress, and our existing platforms support this transition. Again, this ensures participating regions and organizations will take advantage of the "crest of the wave," leading in implementation and related standards definitions.

3. The Solution: FairCOVENANT Smart Trade Hubs w/ SmartFlow

The Smart Trade Hubs initiative integrates physical infrastructure, digital governance, and AI-driven compliance and financial services, enabling safe, verifiable, and inclusive trade channels between cities. We provide access to our SmartFlow supply chain orchestration platform integrated with SmartMarket, a prototype transactions environment designed to support verified cross-border transactions.

Core Components

The FairCovenant Smart Trade Hubs initiative features a deep value stack with a growing number of partners. Spanning from the physical layers rooted in the local economy, to the digital, financial, and people-to-people interactions.

FairReal: Physical Layer (Partners)

- Wholesale display floors for B2B trade showcases, perhaps anchored in industrial and trade zones.
- Warehousing and cold storage for order fulfillment.
- Packaging and logistics services integrated with compliance processes.

[7] See: Rao. S.X. et al. (2025) "Fraud Detection at eBay" in Emerging Markets Review Volume 66, `https://tinyurl.com/57nzxcst`; and `https://tinyurl.com/57bw3fty`.

[8] See OECD (2021), E-Commerce Challenges in Illicit Trade in Fakes: Governance Frameworks and Best Practices, Illicit Trade, OECD Publishing, Paris, https://doi.org/10.1787/40522de9-en. Accessed here: `https://tinyurl.com/27fpn6tx`

[9] See Merchants Risk Council (2025) 2025 Global eCommerce Payments and Fraud Report. Accessed here: `https://rb.gy/b37foh`

- Data center-ready or capable, to support deepening digitalization for connected organizations and enterprises.
- Downstream value-adding and processing capabilities, resulting from more effective aggregation of fragmented upstream MSMEs.

The physical layer is designed to endow MSMEs with the distribution leverage of major retailers. We do so by streamlining the fulfillment process for MSMEs traditionally locked out of cross border trade opportunities due to fixed capital constraints.

These warehousing and logistics centers also act as the focal point for supply chain orchestration, supported by the FairFlow digital ecology. Finally, co-location of FairLearn centers within the trading environment provides opportunities to align applications development with in-situ problem solving, enabling faster prototyping and implementation of solutions in areas such as IOT and robotics. See below.

Fair01: **Digital Layer**

Technical leverage for MSMEs by giving them the tools of total supply chain organization accessible previously only to mega-corps such as JD and Amazon.

Consignment documentation and administration is built-in to the existing FairFlow App, coupled with supply chain traceability and certifications compliance workflows. Transparency and auditability are enabled via the blockchain backbone. Our approach delivers important benefits:

1. Supply Chain digitalization and marketplace for MSMEs to access global buyers without upfront cost. We do this by embedding FairFlow and FairMarket tools on day-1, so that partners can begin their digitalization and market access journey straight away.
2. FairCovenant governance architecture:
   - Provenance tracking for all trade events;
   - Blockchain-tracked data update audits; and
   - AI-powered verification of compliance and certification integrity (e.g., Halal).
3. (pro)Metheus Oracles for real-time data validation and Causal-Intelligence for financiers, insurers, and regulators. (To be developed in collaboration with partners.) [10]
4. Distributed data network with each Smart Trade Hub facility supporting data nodes for blockchain ledgers and localized AI model deployment.

FairLearn: **Talent Layer**

FairLearn Centers are incubators and training centers aimed at enabling talent access for both MSMEs and municipalities under the banner: “Build for the Global South”.

1. On-site developer hub to build:
   - Sector-specific AI applications.
   - Trade finance and compliance solutions.
   - Productivity solutions utilizing IOT and robotics.
2. Upskill local workforce in AI, compliance and digital trade systems.
3. Physical domicile for development teams.

FairExchange: **City-to-City Trade Corridors**

Sister City relationships have been developed across the globe. They have been in place for many decades, but many are limited to periodic cultural exchange activities. We aim to leverage these relationships and transform them into economic development platforms that underpin the Smart Trade Hubs ecosystems.

- Based on sister-city relationships, ensuring mutual trust and governance alignment.
- Existing commitments and resourcing can be harnessed and leveraged.
- Focal locations: we aim to focus on developing nodes in China-Africa-SE Asia, targeting textiles, agri-products, and consumer goods.

4. Operational Model and Revenue Streams

The Smart Trade Hubs initiative monetizes via transaction-based and service-based models, ensuring low barriers for MSMEs while creating high-value recurring revenues from compliance services. We can also leverage fixed capital to generate new revenue streams, distributing this value to all stakeholders so as to drive global access and competitiveness of MSMEs. Targeted and potential revenue stream opportunities are summarized in table (2).

Key Differentiators

The proposed initiative has a number of key differentiators that distinguishes us from the pack. These include:

- **Global credibility**: the team is anchored by globally-recognized leaders in supply chain digitalization research and development, with strong industry, R&D and government support.
- **Existing technology**: leverage existing technologies that have already been developed. This means we are not starting *tabula rasa*, but possess tools ready to implement, so as to fast-track engagement and on-boarding.
- **Global network of MSME**: clusters in multiple countries. This network lays the foundations for scaling.
- **Platform Leverage**: acts as a trade finance and insurance aggregator, potentially spinning out a financial services arm.

[10] See (pro)Metheus white paper: `https://github.com/metheusai/papers/blob/main/prometheus-wp.pdf`

5. Stakeholder Benefits

The project concept delivers benefits to all participating stakeholders. The core benefits for principal beneficiaries are described below.

Municipal/Regional Government (Anchor Partners)

- **Economic Growth**: Boosts exports and attracts investment.
- **MSME Empowerment**: Enables local MSMEs to compete globally.
- **Global Positioning**: City becomes a trusted trade hub.
- **Talent Development**: FairLearn Center creates a digital workforce pipeline.
- **Policy Leadership**: Set standards for AI governance and trade compliance.

MSMEs

- Access to global buyers and sellers via trusted digital channels.
- Verified suppliers and products with transparent data for auditing and validation.
- No upfront fees - pay only when sales occur.
- Simplified compliance and financing.
- Virtual aggregation and value-adding potential.
- Access to trade-enabling finance and insurance services.

Industrial/Trade Zone/Warehousing

- Activate existing infrastructure.
- Differentiate industrial zone/warehouse via access to global networks and streamlined trading solutions.
- Become a focal point for aspiring supply chain orchestrators, value-adders (eg. downstream processors) and associated solutions providers.

Financial Institutions & Providers

- Lower default risk via auditable trade data.
- Access to under-served MSME segments.
- Ability to deploy Halal-compliant products in growing markets.
- Access to Fair01 data validation oracles to enable streamlined payments and settlements against validated data.

Insurers

- Reduced fraud via verified trade events.
- Faster claims resolution with data integrity assurance.
- Automated adjudication process via the (pro)Metheus Causal-Intelligence Oracle.

AI Foundational Model Developers

- Real-world deployment in regulated trade environments.
- Access to sector-specific data and application-layer co-development.
- Brand positioning as trusted AI for compliance-critical use cases.
- Privileged access to high-quality datasets via the Metheus data derivatives market.

Education & Training Sector

- Become a regional hub for technical talent incubation and development.
- Align training with industry demand for AI and compliance skills.
- Access to high quality technical personnel through FairLearn global footprint.

6. Early Investment and Partnership Opportunity

We invite anchor municipal and regional governments, AI partners, organizations involved in trade and MSME curation and orchestration, and financial/insurance institutions to join as foundational contributors to this project.

Municipal/Regional Government Role

- Provide financial support.
- Provide site, base infrastructure and governance leadership.
- Gain economic, branding, and policy advantages.
- Coordinate MSME participation.

Industrial Zone/Trade Zone and Warehouse operators

- Contribute space to support FairLearn Center development.
- Integrate FairFlow and FairMarket systems to support tenant development.

AI Partners (Open Source Model Developers)

- Contribute compute credits, developer teams (3–4 developers and analysts), and financial support.
- Share in downstream IP and commercialization opportunities.

Finance and Insurance Partners

- Deploy pilot trade finance and insurance products.
- Access aggregated, digitally verified MSME trade flows.

7. Roadmap

Phase 1: Market Sounding (now) - Q3 2025

- Determine interest.
- Scope and refine roadmap.

Phase 2: Expressions of Interest and pre-commitments - Q4 2025

- Non-disclosure agreements.
- Terms sheets and letters of intent.
- Finalize operational incorporation etc.

Phase 3 (3–6 months) - Q1 and Q2 2026

- Secure pilot site and anchor city partnership.
- Onboard initial MSMEs.
- Deploy digital infrastructure and begin FairLearn training.
- Design and build (pro)Metheus Oracles.

Phase 4 (6–12 months) - Q2, Q3, Q4 2026

- Operationalize logistics and compliance layers.
- Launch first trade flows with integrated finance and insurance.
- Integrate the (pro)Metheus validation service into production environment, whilst developing the (pro)Metheus Causal-Intelligence insurance service.

Phase 5 (12–24 months) - Q4 2026 onwards

- Achieve corridor-level trade volumes.
- Document blueprint for replication in other city pairs.
- Use outcomes to mobilize regional and global standard-setting bodies.
- Push (pro)Metheus Causal-Intelligence Oracle insurance service into production
- Develop the Metheus data-derivative market, so as to attract additional developer interest in fielding AI-native applications. [11]

Conclusion

The FairCovenant Smart Trade Hubs initiative is a strategic project that combines physical trade infrastructure, digital compliance governance, and AI innovation to unlock global trade for MSMEs while setting a new benchmark for trusted trade corridors.

By joining as an early partner, stakeholders gain first-mover advantage, policy leadership, and access to high-value downstream opportunities in digital trade, AI and Halal markets.

[11] See technical monograph for the Metheus data market: `https://bit.ly/4lTv5T0`

Proponents

This initiative has been developed by Smart Trade Networks and Metheus.

Smart Trade Networks group is organized around Smart Trade Networks Global Limited - a Hong Kong-registered company, with R&D undertaken from its base in Brisbane, Australia.

Smart Trade Networks has delivered award-winning projects in collaboration with industry, universities and government agencies in multiple countries.

The team is led by Dr. Warwick Powell, Adjunct Professor at Queensland University of Technology.

Contact and Coordination

For further information, please contact:

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Warwick is a global leader in supply chain digitalization, payments and orchestration. He is the author of China, Trust and Digital Supply Chains. Dynamics of a Zero Trust World (Routledge, 2023) and a large body of peer-reviewed journal articles.

Table 1: FairCovenant Smart Trade Hubs *Diversified Revenue Streams*

Stream	Source	Model
FairMarket Commissions	MSMEs	Typical range of 1-3% per Transaction (Pay on sale)
Fair01 Data Validation Fees	Banks & Insurers	Unit price charged on a cost+ basis for database access & model training costs, amortized over the number of API calls. [12]
FairPay Finance Brokerage	Mainstream specialized & Islamic banks & Lenders	Referral & Arranger Fees
FairPay tokenized real time settlements	MSMEs	Settlements fee for the provision of the service (% of transaction fee) [13]
FairPay Insurance Brokerage	Global & local insurers	Commission per policy
Fair01 Technology Brokerage & Certification	Technology suppliers	Certification fees & sales fees on FairMarket
FairReal Logistic Services	MSMEs	Storage, packaging, cold chain fees
FairCovenant Analytics	Governments, enterprises	Compliance dashboards & analytics

Table 2: *12. A typical supply chain process involves up to 10-15 main data update events, each with multiple data points required for validation.*
13. Real time tokenized transactions settlement can be supported in jurisdictions where stablecoins are permitted.